Probabilistic Forecasting of Future Market Trends and Volatility Envelopes

Self-Supervised Alpha Trend Distribution Predictor — Project Proposal

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1. PROBLEM AND MOTIVATION

The goal of this project is to predict future market behavior at time $(t+k)$ using only the information available up to time (t). Rather than predicting individual prices or only the direction of the next trend, we aim to forecast multiple future trend segments and their associated volatility envelopes, capturing the range of possible market movements and enabling the estimation of conditional probability distributions of future market behavior. To achieve this, we will experiment with different supervised and self-supervised learning architectures, and compare alternative modeling approaches to identify a model that produces meaningful predictions and generalizes reliably to unseen market periods. This matters because quantitative researchers and traders need not only predictions of market direction but also reliable estimates of the range, uncertainty, and probabilities of future movements to support better-informed trading and risk-management decisions.

Figure 1 summarizes the idea: information from several markets up to time t is mapped to a latent regime space, from which multiple trend segments, volatility envelopes, and non-Gaussian conditional distributions are generated for the future horizons $t+k_1$, $t+k_2$, and $t+k_3$.

2. IMPACT

If successful, this project could help quantitative researchers better understand and predict future market trends by estimating conditional probability distributions and volatility envelopes rather than relying on single-point predictions. Comparing supervised and self-supervised architectures will help determine which approaches capture meaningful market patterns and generalize best to unseen data.

If the models fail to generalize, we will learn about their limitations in capturing market behavior. We will also consider risks such as changing market conditions, extreme events, data bias, and information leakage, paying particular attention to the reliability of predicted probabilities and the consequences of underestimating market risk.

3. ARCHITECTURAL COMPARISON AND SYNTHESIS

The structural properties, theoretical assumptions, and operational trade-offs of the analyzed literature are compared against the proposed framework in Table 1. The compared works are Stock-JEPA and TS-JEPA [1, 2], TNC [3], conditioned normalizing flows and MANF [4, 5], the economics-informed neural SDE ARTEMIS [6], and Market-1T and LaT-PFN [7, 8]. Closely related work includes JEPA learning of latent market states [9] and neural Lévy SDEs for density forecasting [10].

4. DATA

Data Source and Description

This project uses historical stock-market data for SPY, which tracks the S&P 500, and QQQ, which tracks the Nasdaq-100. The preliminary dataset was downloaded using the Python *yfinance* package, which retrieves market data from Yahoo Finance [11] (`https://finance.yahoo.com`).

The dataset contains **501 daily observations for each ETF**, covering October 21, 2024, to October 20, 2026. Each observation includes five numerical features: Open, High, Low, and Close prices, measured in USD per share, and Volume, measured in shares traded. The date serves as the time index. The data were downloaded programmatically and organized chronologically.

The prediction target will be the direction, slope, and duration of the next trend segment, extracted from the price series using ℓ_1 trend filtering [12]. These targets will be constructed during preprocessing rather than obtained directly from the original dataset.

Data Quality and Preprocessing

The initial quality check found **zero missing OHLCV values** for both ETFs. However, the data may contain market noise, extreme returns, and non-stationary behavior. The observed daily returns include large positive and negative movements, which require careful examination. Preprocessing will include checking duplicate dates, validating OHLC values, calculating daily returns, and normalizing features using training data only. Chronological train, validation, and test splits will be used to prevent information leakage.

Exploratory Analysis

Figure 2 shows the daily closing prices of SPY and QQQ over the observation period. Both series exhibit an overall upward movement, interrupted by noticeable price declines and recoveries. Figure 3 presents their daily return distributions, which are concentrated around zero but include more extreme movements. Table 2 summarizes the initial dataset and return statistics.

The preliminary analysis suggests that the dataset is suitable for investigating market trends, while QQQ exhibits greater daily return variability than SPY in this sample. Further analysis will examine the distribution of the trend-segment targets after applying ℓ_1 trend filtering. This ex-

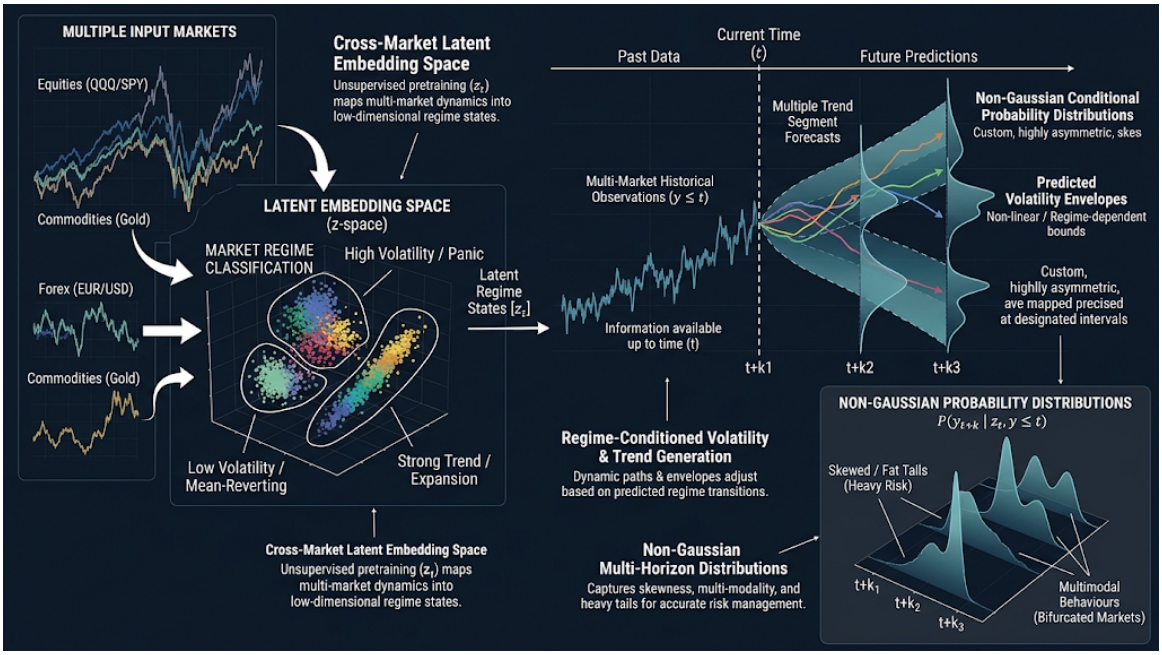


Figure 1: *Overview of the proposed framework. Multiple input markets (equities, commodities, forex) are embedded into a cross-market latent space of regime states, which conditions the generation of multi-horizon trend segments, volatility envelopes, and non-Gaussian probability distributions.*

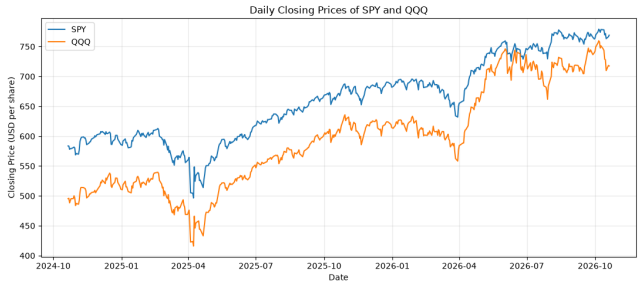


Figure 2: *Daily closing prices of SPY and QQQ (USD per share), October 2024 to October 2026. Both series trend upward, with a sharp decline and recovery around April 2025.*

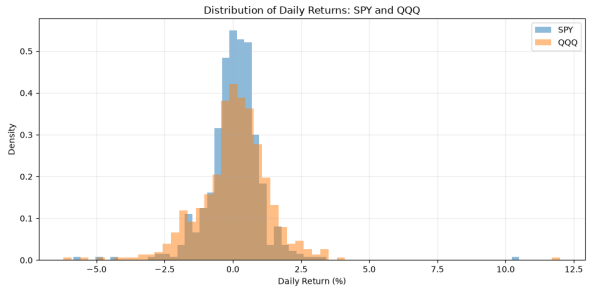


Figure 3: *Distribution of daily returns (%) for SPY and QQQ. Returns are concentrated near zero with heavy tails; QQQ shows wider dispersion than SPY.*

ploratory stage provides an initial assessment of data quality and project feasibility.

5. PROPOSED MODELS AND METHODS

To evaluate representation quality methodically and avoid an abrupt jump from classical baselines to non-contrastive self-supervised architectures, we investigate a five-tier modeling hierarchy. This progression evaluates linear mappings, sequential recurrence, reconstructive feature bottlenecks, contrastive embeddings, and non-reconstructive predictive representations under identical evaluation protocols.

1. **Linear and Logistic Regression Baselines:** Linear regression predicts continuous trend characteristics (such as the slope and duration of the next segment) from engineered OHLCV features and returns, while multinomial logistic regression classifies directional state (upward, downward, or flat). These models establish the primary benchmark to test whether simple linear relationships are sufficient.
2. **Recurrent Sequence Modeling (LSTM):** A Long Short-Term Memory network captures temporal order and nonlinear sequential dynamics across historical windows. This step tests whether modeling path dependency and volatility memory across time offers clear advantages over static linear feature projections.
3. **Reconstructive Latent Modeling (Autoencoder / U-Net):** An encoder-decoder architecture compresses sequential market features into a low-dimensional bottleneck by optimizing input reconstruction. This evaluates whether non-linear dimensionality reduction and manifold compression capture underlying trend dynamics, or whether reconstruction objectives overfit high-frequency market noise.
4. **JEPA-Based Predictive Architecture:** A Joint-Embedding Predictive Architecture [13, 14] learns

representations of historical market contexts and directly forecasts target representations of future segments in latent space. Downstream heads then predict trend parameters, directional classes, and volatility envelopes. This model tests whether predicting abstract future states rather than raw prices or contrastive pairs yields better generalization.

All models will be trained and evaluated using the same chronological data splits and standardized metrics. The intermediate architectures ensure that any performance gains from JEPA can be attributed specifically to non-reconstructive representation learning rather than ordinary non-linearity or temporal modeling.

Preliminary Baseline Results

The dataset has been downloaded and explored, but the models have not yet been trained. Therefore, training and validation metrics will be reported after implementing and running at least one baseline, using the training and validation sets defined in Section 6.

6. EVALUATION

Evaluation Metrics

The model will be evaluated on both predictive precision and practical trading performance using three complementary metrics:

- **Profit and Loss (PnL):** cumulative strategy returns net of transaction costs and slippage, to evaluate practical financial viability.
- **Maximum Drawdown (MDD):** the largest peak-to-trough decline in cumulative equity, penalizing unstable return profiles.
- **Directional Accuracy:** the proportion of correctly predicted trend-segment directions (upward, downward, or flat), to assess raw forecasting quality.

Data Splitting and Experimental Protocol

The dataset is split chronologically into three non-overlapping subsets (Table 3).

To eliminate lookahead bias, strict chronological ordering is maintained without random shuffling. Normalization parameters and feature selections are calculated strictly on the training split. Segment labels and forecasting horizons near subset boundaries are truncated, in the spirit of purging and embargo [15], to prevent target leakage, and the test set remains untouched until final evaluation.

Baseline Comparison

The JEPA-based model will be benchmarked against a polynomial regression model fitted on historical market indica-

Table 1: *Comparison of related architectures and the proposed framework.*

Model / Framework	Architectural Paradigm	Latent Space & Regime Representation	Distributional Formulation	Multi-Horizon & Volatility Capability	Primary Methodological Limitations Addressed by Proposed Project
Stock-JEPA / TS-JEPA	Self-Supervised Latent Predictive Architecture (EMA target encoder)	Latent anchor conditioned on fixed financial prior statistics; single-market or equity cross-section	Deterministic latent projections; no explicit non-Gaussian density estimation	Fixed multi-step scalar return predictions; lacks continuous volatility envelopes	Bypasses generative noise reconstruction, but cannot output calibrated prediction intervals, tail distributions, or multimodality.
TNC (Tonekaboni et al.)	Contrastive Learning via Temporal Neighborhood Coding	Unsupervised state clustering bounded by ADF local stationarity tests	Non-probabilistic; latent distances optimized via contrastive loss	Single-step state assignment; purely descriptive historical clustering	Backward-looking and descriptive; heuristic stationarity tests fail during market jumps; unable to project forward out-of-sample uncertainty.
Conditioned Normalizing Flows (Rasul et al. / MANF)	Autoregressive Sequential Backbone + Invertible Generative Flow	Observation-space sequential representations (RNN / Attention); lacks explicit regime disentanglement	Non-Gaussian, exact log-likelihood via bijective transformations (Real NVP)	Autoregressive multi-horizon rollouts; susceptible to error accumulation	Autoregressive feedback loop causes error cascades in volatile financial regimes; vulnerable to Jacobian instability during market shocks.
Economics-Informed Neural SDEs (ARTEMIS)	Continuous-Time Stochastic Calculus (Drift, Diffusion, Jumps)	Microstructural state space with Feynman–Kac and Sharpe ratio regularization	Continuous state-dependent diffusion and compound Poisson jump processes	Continuous-time trajectory sampling and parametric prediction intervals	Conflates single-asset microstructural noise with genuine volatility shifts; computationally expensive continuous path integration.
Market-1T / LaT-PFN	Cross-Asset Foundation Embeddings / In-Context Latent JEPA	Unified latent world model spanning cross-market assets and heterogeneous regimes	Zero-shot predictive distributions / Linear probe feature representations	In-context multi-horizon evaluation; downstream linear classification and ranking	Restricts downstream evaluation to static linear probing or single-step portfolio sorting; lacks calibrated non-linear volatility envelopes.
Proposed Framework	**Hybrid Cross-Market JEPA + Conditioned Non-Gaussian Flow & Neural SDE**	**Cross-market latent manifold z_t capturing macro regimes without raw input reconstruction**	*Targets multi-horizon trend segments, volatility envelopes, and calibrated non-Gaussian distributions (Sections 5 and 6).*		

Table 2: *Summary of the dataset and daily return statistics.*

Statistic	SPY	QQQ
Daily observations	501	501
Missing OHLCV values	0	0
Mean daily return	0.0606%	0.0835%
Daily return std. dev.	1.0416%	1.3755%
Minimum daily return	−5.8543%	−6.2109%
Maximum daily return	10.5019%	12.0031%

Table 3: *Chronological train / validation / test split.*

Subset	Share	Purpose
Training set	75%	Fit model parameters
Validation set	15%	Tune hyperparameters and configurations
Test set	10%	Benchmark final out-of-sample performance

tors. Both models will use identical chronological splits, input features, target boundaries, and evaluation metrics. The comparison focuses on whether JEPA's learned latent representations yield lower forecast errors, higher directional accuracy, and superior net financial returns compared to a standard regression baseline.

Additional Experiments and Robustness

- **Ablation Study:** a comparative run using raw OHLCV features directly in the prediction head (omitting the JEPA encoder) will isolate the value added by the self-supervised representations.
- **Regime Sensitivity:** performance will be disaggregated into calm and high-volatility regimes to test cross-environment stability.

The core hypothesis will only be considered supported if the JEPA model achieves consistent out-of-sample gains over the baseline; positive PnL accompanied by severe drawdowns or unreliable directional accuracy will not be treated as success.

7. RESOURCES AND CODE

The project will initially be developed and tested on a regular laptop using Python and its data-science libraries. **No GPU or AUB HPC cluster is required at the initial stage.** If training the JEPA model becomes computationally expensive, additional computing resources may be considered.

The project code, data-loading scripts, baseline models, and setup instructions will be maintained in a public GitHub repository: `https://github.com/CHCICH/Self_Supervised_Alpha_Trend_Distribution_Perdictor`. Regular commits will be made throughout the semester to document the project's progress and development.

8. TIMELINE

Table 4: *Week-by-week plan to the final presentation.*

Period	Tasks	Who
Oct. 21–27	Finalize data collection and preprocessing; complete exploratory analysis.	All
Oct. 28–Nov. 3	Implement and evaluate linear and logistic regression baselines.	All
Nov. 4–10	Develop the other models and compare preliminary results.	All
Nov. 11–17	Analyze results and prepare the progress report (due Nov. 17).	All
Nov. 18–24	Improve the models, run additional experiments, and analyze different market conditions.	All
Nov. 25–Dec. 3	Finalize results, prepare the poster, and complete the presentation.	All
Dec. 4–10	Present the project and complete the final report.	All

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